

Environmental Monitoring with AI — Project Documentation

1. Overview

This project demonstrates a toy remote-sensing workflow for exoplanets. It mimics how astronomers infer:

- Approximate planet equilibrium temperature from star and orbit parameters (thermal-IR proxy).
 - Presence of atmospheric water vapor from spectral feature strength and observability proxies.
- It builds a labeled dataset, trains a classifier, evaluates performance, and predicts whether water vapor is “detected” for new targets. Educational use only; not a rigorous scientific pipeline.

2. Key Features

- Computes equilibrium temperature (T_{eq}) from stellar temperature, stellar radius, orbital distance, and albedo.
- Observational proxies:
 - Transit depth (ppm)
 - Water absorption feature strength (ppm)
 - Phase curve amplitude (ppm)
- Generates a “water_detected” label via a simple rule.
- Trains a Decision Tree classifier and reports metrics.
- Example prediction for a Sun–Earth analog.

3. Data Requirements

Required CSV columns:

- `koi_steff`: Stellar effective temperature (K)
- `koi_srad`: Stellar radius (solar radii)
- `koi_sma`: Semi-major axis (AU)
- `koi_prad`: Planet radius (Earth radii)

Optional (simulated if absent):

- `albedo` (0–1; default 0.3)
- `transit_depth_ppm` (ppm)
- `h2o_feature_depth` (ppm)
- `phase_amp_ppm` (ppm)

Place `exoplanet_data.csv` alongside the script or adjust the path.

4. Installation

- Python 3.9+
- pip install numpy pandas scikit-learn

5. How It Works

5.1 Load data:

- Reads CSV, applies safe defaults for optional columns.

5.2 Derived quantities:

- Unit conversions.
- $T_{eq} = T_{star} \cdot \sqrt{R_{star} / (2a)} \cdot (1 - A)^{(1/4)}$
- Transit depth $\approx (R_p/R_s)^2 \times 1e6$ ppm if missing.
- Simulates h2o_feature_depth and phase_amp_ppm if missing.

5.3 Label ("water_detected"):

- snr_ok: transit_depth_ppm ≥ 200 or phase_amp_ppm ≥ 50
- feat_ok: h2o_feature_depth ≥ 20
- temp_ok: $250 \leq T_{eq} \leq 1200$ K
- water_detected = 1 if all true; else 0

5.4 Model:

- Features: [Teq, koi_prad, koi_sma, koi_steff, transit_depth_ppm, h2o_feature_depth, phase_amp_ppm, albedo]
- DecisionTreeClassifier(max_depth=4, random_state=42)
- 80/20 train/test (stratified if possible)

5.5 Outputs:

- Accuracy, class balance, classification report.
- First 10 predictions vs. actual.
- Example prediction and confidence (if available).

6. Usage

- Save script as exo_remote_sensing.py.
- Ensure exoplanet_data.csv is present.
- Run: python exo_remote_sensing.py
Outputs include metrics, sample predictions, and an example target result.

7. Configuration

- Adjust thresholds in the detection rule (transit_depth_ppm, h2o_feature_depth, Teq window).
- Try other models (LogisticRegression, RandomForest, Gradient Boosting).
- Use cross-validation for robustness.

8. Example I/O

Minimal CSV columns example:

- `koi_steff`: 5778; `koi_srad`: 1.0; `koi_sma`: 1.0; `koi_prad`: 1.0

Example prediction inputs:

- `h2o_feature_depth`: 25 ppm; `phase_amp_ppm`: 40 ppm

Note: Earth–Sun transit depth ~84 ppm; `snr_ok` may be false unless `phase_amp_ppm` is increased.

9. File Structure

- `exo_remote_sensing.py` — main script
- `exoplanet_data.csv` — dataset

10. Limitations

- Synthetic labels; the model reproduces a designed rule.
- Simplified Teq; ignores greenhouse, eccentricity, redistribution.
- Simulated features if missing; not suitable for real detection claims.
- No surface water mapping; only atmospheric vapor detectability proxy.

11. Extending the Project

- Replace `h2o_feature_depth` with real spectral retrieval measurements (e.g., 1.4 μm band).
- Include emission spectra and brightness temperatures for hot planets.
- Add uncertainties/SNR propagation; consider probabilistic labels.
- Better evaluation: cross-validation, ROC/PR curves, calibration.
- Inspect feature importance (`clf.feature_importances_`).

12. Troubleshooting

- Missing columns: rename CSV headers or adjust `safe_col()` names in the script.
- Many NaNs dropped: ensure `koi_srad`, `koi_steff`, `koi_sma`, `koi_prad` are populated and units are correct.
- All labels 0 or 1: tune thresholds.
- Low accuracy: align thresholds with your simulated/measured signals.

13. References

- Equilibrium temperature via radiative balance.
- Transit depth: $(R_p/R_s)^2$.
- Remote sensing: transmission/emission spectroscopy, phase curves, high-contrast imaging.
- Libraries: pandas, scikit-learn.

14. License

- Educational sample. Add an open-source license (e.g., MIT) if sharing.

15. Contact

- Share your CSV schema or a sample row to tailor the script to your dataset.